

▪ Problem 3 (05/02/2021)

A steel planar surface with a length equal to 500 mm and a width equal to 50 mm must be flattened. A face milling roughing operation will remove a machining allowance equal to 5 mm. The face mill will be positioned so that 10% of the tool diameter will exit from one side of the plane width.

Determine:

- a) The cutting torque and the cutting power, verifying that the operation can be carried out with a milling machine with a power of 12 kW and an efficiency of 0,8.
- b) The cutting time needed to carry out the operation.

Data: mill diameter 80 mm, number of cutting edges $Z = 7$, entering angle = 90° , $f_z = 0,15$ mm/rev, $k_{cs} = 1600$ MPa, $x = 0,197$.

Cutting speed:

if $f_z = 0,05$ mm/rev then $v_c = 305$ m/min;

if $f_z = 0,1$ mm/rev then $v_c = 275$ m/min;

if $f_z = 0,2$ mm/rev then $v_c = 225$ m/min.

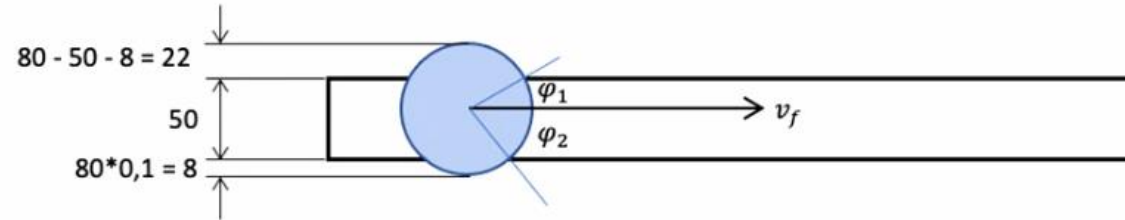


▪ **Solution 1**

- a) *The cutting torque and the cutting power, verifying that the operation can be carried out with a milling machine with a power of 12 kW and an efficiency of 0,8.*

In order to answer, it is necessary to understand what the actual situation in this face milling operation is.

First of all, we need to compute the engagement angle and then the number of simultaneously working teeth.



Referring to the figure, the overall engagement angle is the sum of the following two angles:

$$\varphi_1 = \arcsin\left(\frac{80/2 - 22}{80/2}\right) = 26,74^\circ$$

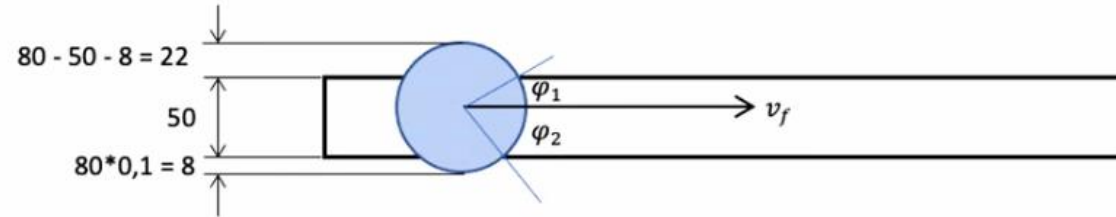
$$\varphi_2 = \arcsin\left(-\frac{80/2 - 8}{80/2}\right) = -53,13^\circ$$

$$\varphi = \varphi_1 + |\varphi_2| = 26,74^\circ + 53,13^\circ = 79,87^\circ$$

$$\varphi_0 = \frac{2\pi}{Z} = \frac{360}{7} = 51,43^\circ$$

$$z = \frac{\varphi}{\varphi_0} = \frac{79,87^\circ}{51,43^\circ} = 1,55$$

▪ **Solution 1**



Therefore, we have to consider the worst-case scenario when two cutting edges are simultaneously cutting, that we have when the angular position of the two cutting edge is symmetric with respect to the feed direction, so:

Cutting tooth 1, angular position: $\theta_1 = \frac{\varphi_0}{2} = \frac{51,43^\circ}{2} = 25,715^\circ \leq 26,74^\circ$

Cutting tooth 2, angular position: $\theta_2 = -\frac{\varphi_0}{2} = -\frac{51,43^\circ}{2} = -25,715^\circ \geq -53,13^\circ$

The two angular positions are coherent with respect to the engagement angle.

In order to do answer the question we notice that:

$$h_D(\theta_1) = f_Z \cos \theta_1 = 0,15 \cdot \cos 25,715^\circ = 0,135 \text{ mm}$$

$$h_D(\theta_2) = f_Z \cos \theta_2 = 0,15 \cdot \cos -25,715^\circ = 0,135 \text{ mm}$$



▪ **Solution 1**

Therefore:

$$h_D(\theta_1) = h_D(\theta_2) = h_D$$

It follows:

$$A_D = h_D \cdot a_p = 0,135 \cdot 5 = 0,675 \text{ mm}^2$$

$$k_c = k_{cs} h_D^{-x} = 1.600 \cdot 0,135^{-0,197} = 2.374 \text{ MPa}$$

$$F_c = k_c A_D = 2.374 \cdot 0,675 = 1.602 \text{ N}$$

$$T_c = F_{c,\theta_1} \frac{D/2}{1000} + F_{c,\theta_2} \frac{D/2}{1000} = 2 \cdot F_c \frac{D/2}{1000} = 2 \cdot 1.602 \frac{80/2}{1000} = 128,2 \text{ Nm}$$

Being $f_z = 0,15$, from the data by linear interpolation we can choose the adequate cutting speed as:

$$v_c = \frac{275 + 225}{2} = 250 \frac{\text{m}}{\text{min}}$$

So:

$$\omega = \frac{1000}{60} \cdot \frac{v_c}{D/2} = \frac{1000}{60} \cdot \frac{250}{80/2} = 104,2 \frac{\text{rad}}{\text{s}}$$

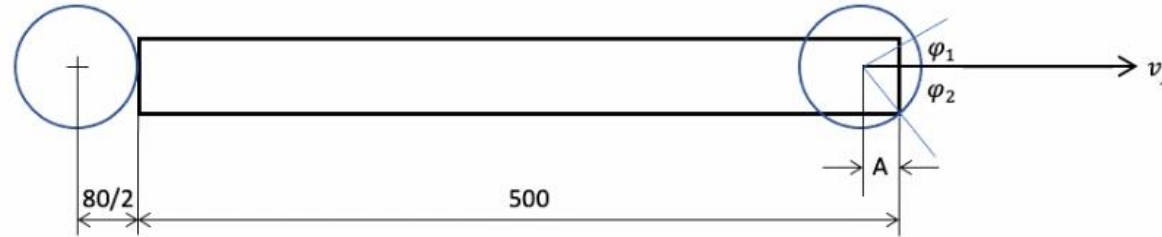
$$P_c = T_c \omega = 128,2 \cdot 104,2 = 13.358 \text{ W} = 13,4 \text{ kW}$$

$$P_{c,limit} = E \cdot P_e = 0,8 \cdot 12 = 9,6 \text{ kW}$$

Therefore, being $P_c = 13,4 \text{ kW} \geq P_{c,limit} = 9,6 \text{ kW}$, the face milling operation is not feasible.



▪ **Solution 2**



b) *The cutting time needed to carry out the operation.*

Referring to the figure, the cutting time is equal to:

$$t_c = \frac{L_{workpiece} + \frac{D}{2} - A}{v_f} = \frac{500 + \frac{80}{2} - A}{v_f}$$

Where

$$A = \min \left\{ \frac{D}{2} \cos \varphi_1; \frac{D}{2} \cos \varphi_2 \right\} = \min \left\{ \frac{80}{2} \cos 26,74^\circ; \frac{80}{2} \cos 53,13^\circ \right\} = \min\{35,72; 24\}$$

$$= 24 \text{ mm}$$

$$n = \frac{1000 \cdot v_c}{\pi D} = \frac{1000 \cdot 250}{\pi \cdot 80} = 995 \text{ RPM}$$

$$v_f = Z \cdot f_z \cdot n = 7 \cdot 0,15 \cdot 995 = 1.045 \frac{\text{mm}}{\text{min}}$$

Therefore:

$$t_c = \frac{500 + \frac{80}{2} - 24}{1.045} = 0,494 \text{ min} = 29,6 \text{ s}$$

